DEPARTMENT OF INFORMATION TECHNOLOGY
COMPETITIVE CODING
23 BATCH V SEM IT-2, B-1 AND IT-3 B-1

GREEDY ALGORITHMS

1. 252. Meeting Rooms (Easy)
Given intervals of meeting times [start, end], determine if a person can attend all meetings (no overlap).
→ *Return True/False.*

2. 253. Meeting Rooms II (Medium)
Given meeting intervals, find the **minimum number of conference rooms** required.
→ *Use sorting + min heap.*

3. 455. Assign Cookies (Easy)
Each child has a greed factor, each cookie has a size. Assign cookies to maximize the number of satisfied children.
→ *Greedy sort and assign.*

4. 452. Minimum Number of Arrows to Burst Balloons (Medium)
Given intervals representing balloons, find the minimum arrows needed to burst all balloons.
→ *Greedy interval merging.*

5. 860. Lemonade Change (Easy)
Each customer pays with $5, $10, or $20. You must provide change using available bills.
→ *Simulate with counters.*

6. 55. Jump Game (Medium)
Given an array of jump lengths, determine if you can reach the last index.
→ *Greedy track farthest reachable.*

RECURSION & BACKTRACKING

7. 78. Subsets (Easy)
Generate all possible subsets (the power set) of a given integer array.
→ *Use recursion/backtracking.*

8. 90. Subsets II (Medium)
Return all subsets for an array that may contain duplicates (no repeated subsets).
→ *Backtrack + skip duplicates.*

9. 39. Combination Sum (Medium)
Find all unique combinations where numbers sum to target. Each number may be reused.
→ *DFS + backtrack.*

10. 46. Permutations (Medium)
Return all possible permutations of a list of distinct integers.
→ *Recursive backtracking.*

11. 47. Permutations II (Medium)
Return all unique permutations for a list with duplicate numbers.
→ *Backtracking + set or visited array.*

12. 131. Palindrome Partitioning (Medium)
Partition a string into substrings such that every substring is a palindrome.
→ *DFS and palindrome check.*
13. 139. Word Break (Medium)
Given a string and a dictionary, determine if it can be segmented into valid words.
→ *Recursion + memoization / DP.*

BINARY TREES & BST

14. 94. Binary Tree Inorder Traversal (Easy)
Return the inorder traversal of a binary tree using recursion or stack.
15. 144. Binary Tree Preorder Traversal (Easy)
Return preorder traversal of binary tree nodes.
16. 145. Binary Tree Postorder Traversal (Easy)
Return postorder traversal of binary tree nodes.
17. 102. Binary Tree Level Order Traversal (Medium)
Return level-by-level traversal using queue (BFS).
18. 103. Binary Tree Zigzag Level Order Traversal (Medium)
Level order traversal but alternate left→right and right→left per level.
19. 199. Binary Tree Right Side View (Medium)
Return nodes visible from the right side of the tree.
20. 543. Diameter of Binary Tree (Easy)
Return the length of the longest path between any two nodes.
21. 110. Balanced Binary Tree (Easy)
Check if the tree is height-balanced (height difference ≤1).
22. 236. Lowest Common Ancestor of a Binary Tree (Medium)
Find the lowest common ancestor (LCA) of two nodes.
23. 100. Same Tree (Easy)
Check if two binary trees are identical in structure and value.
24. 98. Validate Binary Search Tree (Medium)
Check whether a binary tree follows BST properties.

GRAPHS

25. 547. Number of Provinces (Medium)
Given adjacency matrix, count connected components (provinces).
→ *Use DFS or BFS.*
26. 994. Rotting Oranges (Medium)
Each cell is fresh, rotten, or empty. Every minute, rotten oranges infect adjacent fresh ones.
→ *Use BFS to count minutes.*
27. 207. Course Schedule (Medium)
Given courses and prerequisites, determine if you can finish all courses.
→ *Topological sort using Kahn's Algorithm.*
28. 210. Course Schedule II (Medium)
Return a possible order to finish courses given prerequisites.
→ *Topological ordering.*
29. 133. Clone Graph (Medium)
Clone a connected undirected graph where each node contains a value and list of neighbors.
→ *DFS/BFS clone map.*
30. 695. Max Area of Island (Medium)
Given a grid of 0s and 1s, find the size of the largest island (connected 1s).
→ *DFS/BFS flood fill.*